

RGIPT

Subject: Physics

Class: 11

Time Allowed: 45 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class: 11 Section: _____

Section A: Multiple Choice Questions

Attempt all. Each question carries 1 mark.

1. [Easy] 1. A particle starts from rest and accelerates uniformly at 2 m/s^2 for 5 s. What is its final velocity?
A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s [1 Marks]
2. [Easy] 2. The torque produced by a force of 10 N applied at a distance of 0.5 m making an angle of 90° with the lever arm is: A) $5 \text{ N}\cdot\text{m}$ B) $10 \text{ N}\cdot\text{m}$ C) $15 \text{ N}\cdot\text{m}$ D) $20 \text{ N}\cdot\text{m}$ [1 Marks]
3. [Moderate] 3. In rotational motion, angular displacement θ (in rad) is related to linear displacement s (in m) by: A) $s = r\theta$ B) $s = \theta/r$ C) $s = r/\theta$ D) $s = \theta^2 r$ [1 Marks]
4. [Moderate] 4. For a body moving with constant angular acceleration α , the angular velocity after time t is given by: A) $\omega = \omega_0 + \alpha t^2$ B) $\omega = \omega_0 + \alpha t$ C) $\omega = \omega_0 - \alpha t$ D) $\omega = \alpha t$ [1 Marks]

Section B: Short Answer Questions

Answer briefly. Each question carries 2 marks.

1. [Moderate] 1. Derive the linear kinematic equation $s = ut + \frac{1}{2}at^2$ starting from $v = u + at$ and the definition of average velocity. [2 Marks]
2. [Challenging] 2. Explain why torque is a vector quantity and state the right-hand rule for its direction. [2 Marks]
3. [Easy] 3. A wheel of radius 0.3 m rotates with an angular speed of 10 rad/s. Calculate the linear speed of a point on its rim. [2 Marks]

End of Question Paper

Answer Key:

1. B

2. A

3. A

4. B

5. From $v = u + at$, average velocity $\bar{v} = (u + v)/2 = (u + (u + at))/2 = u + \frac{1}{2}at$. Displacement $s = \bar{v} t = (u + \frac{1}{2}at) t = ut + \frac{1}{2}at^2$.

6. Torque $\tau = r \times F$, the cross product of position vector r and force F , giving a vector perpendicular to the plane of r and F . The right-hand rule: curl fingers of right hand from r to F ; thumb points in the direction of τ .

7. Linear speed $v = r \omega = 0.3 \text{ m} \times 10 \text{ rad/s} = 3.0 \text{ m/s}$.